Refbank: an open repository of iterated reference game data

Veronica Boyce[1,2], Alvin W. M. Tan[1], assorted middle authors go here[1], Michael C. Frank[1], & Robert D. Hawkins[1]

[1] Stanford University
[2] Massachusetts Institute of Technology

Author Note

TODO

Correspondence concerning this article should be addressed to Veronica Boyce. E-mail: vboyce@mit.edu

Abstract

[revisit and fix later] Language is an extensible form of communication that can be used to talk about even novel referents. Even when do canonical name exists, people rapidly form shared object-meaning mappings when they have repeated need to communicate about new things. The formation and adaptation of labels for initially novel referents is frequently studied in iterated reference games where a group of people communicate with each other about the same set of potential referents over a series of rounds. While iterated reference games are a common paradigm, the research questions and pre-processing and analysis approaches differ paper to paper so the results cannot be quantitatively synthesized. However, the text and meta-data itself is rich and has high value for re-analysis. We collect iterated reference game data from different papers together into a harmonized repository: Refbank. Refbank is a versioned repository containing transcripts and meta-data from 19 datasets. We explain the process for adding datasets to refbank, what the schema is and what information is stored, how researchers can access refbank, and we show some example analyses on Refbank data. Refbank allows for larger-scale analyses, the study of moderators, and training on reference game data.

Keywords: TODO

Word count: TODO

Refbank: an open repository of iterated reference game data

[TODO ugh I don't like writing pre-intro hooks] Language is an extensible communication system that can be used to communicate about new things even when there are not established terms for reference. The flexible use of language is often studied using iterated reference games, which represent a controlled way of studying how people communicate in the absence of codified labels, and how conventionalized labels form over repeated reference to the same targets. The simplicity and richness of the paradigm has made it a common tool for studying dialogue, convention formation, and now AI systems. The diversity of research topics means that many datasets could be reanalyzed to answer more questions.

Iterated reference games are a commonly used paradigm for studying adaptive language use. A *reference game* consists of a group of people communicating with each other about a set of potential referents. One person, the *describer*, has a specific designated target (usually pre-identified as the target by the experimenter) that they seek to communicate to the *matcher* (or matchers). This act of reference requires coordination between describer and matcher on a symbol–meaning alignment. *Iterated* reference games involve repeated reference to the same targets within the same group of people, so that a certain conversational history builds up that could shape the construction and understanding of later symbols.

Iterated reference games were introduced by Krauss and colleagues in the 1960s (Krauss & Bricker, 1967; Krauss & Weinheimer, 1964, 1966) and popularized by Clark and colleagues in the 1980s (Clark & Wilkes-Gibbs, 1986). This paradigm has been used to study a wide variety of communicative and linguistic phenomena. The primary targets of study are coordination and the formation of conventions or conceptual pacts (Brennan & Clark, 1996; Dahan, 2025; Eliav et al., 2023; Ghaleb et al., 2024; Hawkins, Frank, et al., 2020; Metzing & Brennan, 2003, among others). Some studies focus particularly on iterated reference games as a way of studying collaboration in natural conversations (Clark & Wilkes-Gibbs, 1986; Dahan, 2024; Schober & Clark, 1989; Wilkes-Gibbs & Clark, 1992). Other studies focus on other aspects of coordination, including the negotiation of referring expressions (Bangerter et al., 2020; Knutsen & Le Bigot, 2018) and the role of feedback (Fox Tree & Clark, 2013; Krauss & Glucksberg, 1977; Krauss & Weinheimer, 1966). Iterated reference games with multiple partners have been used to study how speakers design their utterances for their specific partners (Horton & Gerrig, 2002, 2005; Turner & Knutsen, 2021; Yoon & Brown-Schmidt, 2014, 2019a, 2019b) and how speakers might generalize patterns of social conventions across interlocutors (Hawkins, Goodman, et al., 2020; Hawkins et al., 2021). While a majority of iterated reference game studies have focused on typical adult populations, they have also been used with children and teenagers (Branigan et al., 2016; Leung et al., 2024; Turner & Knutsen, 2021) and with older adults (Horton & Spieler, 2007; Hupet et al., 1993; Lysander & Horton, 2012) to study cognitive and linguistic abilities across the lifespan.

A few common patterns recur across papers – participants tend to have high and increasing performance at selecting the correct targets, even as the amount of language produced decreases over repetition. There are also qualitative shifts in the type of language produced, as the descriptions seem to become more conventionalized, sometimes indexed as a transition to more abstract descriptions, and more definite and bare nouns (Brown-Schmidt & Duff, 2016; Clark & Wilkes-Gibbs, 1986; Dahan, 2025; Isaacs & Clark, 1987).

As reviewed above, different researchers have used iterated reference games with different setups and settings as a measure of communication. The breadth of studies in this domain presents an opportunity for evidence synthesis to understand various phenomena related to convention formation across different samples and experimental designs. In particular, there are a number of dimensions of variation in experimental design across such games, including different stimuli choices, different numbers of options, and different sizes and structures of groups. Running a new experiment across all these options would be expensive, so aggregating existing data provides a way to increase what we can learn from the data we already have, and allow for more pinpointed studies in the future (Koile & Cristia, 2021).

Systematic reviews are one possible method for evidence synthesis, and can be helpful to map the conceptual and methodological space (Bovet et al., 2023); however, they do not offer quantitative aggregation of results. Another common approach for evidence synthesis is meta-analysis; unfortunately, the iterated reference game literature is not ideal for meta-analysis because studies report results for different questions and use different measures for the same questions. Additionally, language data of the kind generated in iterated reference games are very rich and amenable to many different possible analyses, which may not be captured by simple statistical aggregation.

A third form of aggregation is to collect raw data from similar paradigms and align them in a shared framework for mega-analysis. This approach enables the reuse of high value data by putting it all in one place in a common format and making it easily accessible. The mega-analytic approach has been used to pool data of various types in the language acquisition space, including transcripts of child–parent interactions (CHILDES, MacWhinney, 2000), parent reports of early vocabulary (wordbank, Frank et al., 2017), and data from the looking-while-listening paradigm (peekbank, Zettersten et al., 2022). Similar approaches are used in neuroscience such as in the OpenNeuro repository (Markiewicz et al., 2021). [TODO are there other examples we should include?]

One advantage of harmonized dataset repositories is that new analytic approaches can be used on the data, including approaches that may not have been available when some data was originally collected (e.g., semantic vector embeddings), or approaches that rely on cross-study data (Frank et al., 2021).

Second, mega-analysis retains individual- and dyad-level patterns; this feature allows for the analysis of individual differences, as well as generative and computational approaches to understanding referential communication in iterated reference games, which require raw data to compare against. For example, iterated reference games can be used to assess the extent to which language models are able to communicate like humans, and actual trial-by-trial data are the most useful for such comparisons [CITE].

Third, data aggregation allows for data reuse for purposes other than analysis. In particular, iterated reference game data can serve as a resource for future experiment design – they can shed light on what aspects of design may be more or less impactful, and may also allow for the selection of stimuli that have desirable properties. Aggregation can also highlight underexplored regions of the experiment design space, suggesting new avenues for iterated reference game research.

Iterated reference games have been widely used to study convention formation, with each experiment manipulating a small set of possible experimental design variables.

Nevertheless, the underlying language data (and associated metadata) from iterated reference games constitute a high value corpus for studying not only convention formation, but also audience design, metaphor, semantic and syntactic alignment, and referring expressions; they can also be used as a benchmark for computational models of pragmatics.

```
## Please visit the URL below to authenticate with your Redivis account:
## https://redivis.com/oauth/authorize?user_code=3ec311089592108310356a2ed114c24f
## Authentication was successful!
```

To enable to reuse of iterated reference game datasets, we present Refbank, an open repository of iterated reference game data. Refbank currently has data from 19 datasets and 8,098 participants (Table 2). In the following sections, we explain the schema of Refbank and what information it contains, describe how researchers can access the data, and show a few analyses on Refbank data to demonstrate its utility.

Design and technical approach

Refbank was constructed to align with the FAIR principles for domain-specific data repositories: findability, accessibility, interoperability, and reusability (**wilkonson2016?**). It is structured as a relational database with different tables encoding different types of data. Because iterated reference games can vary from each other in different ways, we needed a schema that could accommodate different set-ups while still unifying the information for cross-dataset comparison.

A first question is how to define an iterated reference game. For Refbank purposes, we consider reference games where at least some of the stimuli are repeated (in at least some conditions), and change across repetition is of interest. The describer must be a human participant (not an artificial intelligence agent or a confederate). Natural language must be used to communicate.

Core tables

The core information in Refbank is stored in three tables: Trials, Messages, and Choices. The key unit in our schema is the trial, which we define as the actions associated with picking out one target image out of a pool of options. Each trial can be associated with messages (spoken utterances or typed messages from the describer and matchers) and choices (which matcher selected which option).

Trials. Each row in the Trials table contains information about a single trial including what was happening that trial and how it fits into the larger structure of a game.

Each trial is associated with a describer and a list of matchers via their player IDs. Each trial has a target image (seen by the describer) and a set of image options (that matchers choose among), linked via their image IDs. Each trial is labelled as either an "order" trial or a "match" trial. In matching trials, a given target is specifically highlighted and the matcher picks one and then moves on. In ordering, the goal is to reproduce a given order of the set (or subset) of targets. We separate these into trials, but there is flexibility in that a matcher might revise their choice to one trial after later trials.

The basic structure of an iterated reference game is a series of trials (acts of picking out different targets) that are nested within repetitions through the target set (sometimes called blocks or rounds).

The Trials table also contains information about the games that trials are nested within. Each game is a set of people who play with each other, possibly on a network of interlocking pairs or with substitutions (see metadata). Following Hawkins et al. (2021), we include a room number for when there are two trials going on in parallel in the same game (this occurs on network games where two pairs play and then swap).

If there is a change in how the game is working, such as a swap in partners, or in the pool of stimuli, we define that as a new stage, recorded via the stage number. For instance, the network swap games have multiple stages, partner swap designs have multiple stages, and some games with entrainment and test phases are multi-stage and games with a separate "generalization" round have multiple stages. In general, restricting to only "stage 1 data" (as is possible in the visualization app and via filtering downloaded data) allows for a more even comparison across games.

While the trials table includes information about the setup of the trial, the actual content of trial consists of two kinds of actions: messages and choices.

Messages. Each message is one turn of dialogue or one message that was typed in a chat window, depending on the game modality. A message has text, the content of the message, and is associated with a player ID, who produced the message. Messages are also associated with the player's role (either matcher or describer). All messages have a trial ID (linking to the trials table). Off-topic messages such as chit chat or meta-discussion about strategy are included, but marked via a message irrelevant column, to allow for filtering. There is also a time stamp field for when the timing of a message within a trial is known. We do not remove off-topic messages, or trim messages with some extraneous content, but we do use spelling-corrected versions of written messages where available.

Choices. Player's selections are recorded in a choices table. Each choice is tied to a given trial and a given player ID. Each choice is associated with a selected target denoted with choice ID that matches the image ID (and can be compared with the trials target ID). There is again the option for a time stamp to specify how long into the trial the selection was made.

In order games, we only record the final choices players made for each position.

Metadata

In addition to the core information about what happened during the game, we have several tables for metadata about the players, images, and experimental features of the study.

Datasets. The datasets table contains only bibliographic information, linking our dataset ID to the full and short citations for that dataset.

Conditions. Iterated reference games can vary along a number of design features. Most of these factors are coded in the "conditions" table. We code several metadata factors as condition information. These include:

- language: the language in which the iterated reference game was conducted
- group-size: the number of players in a game, where we define game as a set of players who interact with each other but no one else
- partner constancy: whether or not the same set of people is active the entire game. If there are changes in who is interacting, there is not partner constancy.
- role constancy: roles are constant if each player is only the describer or a matcher when

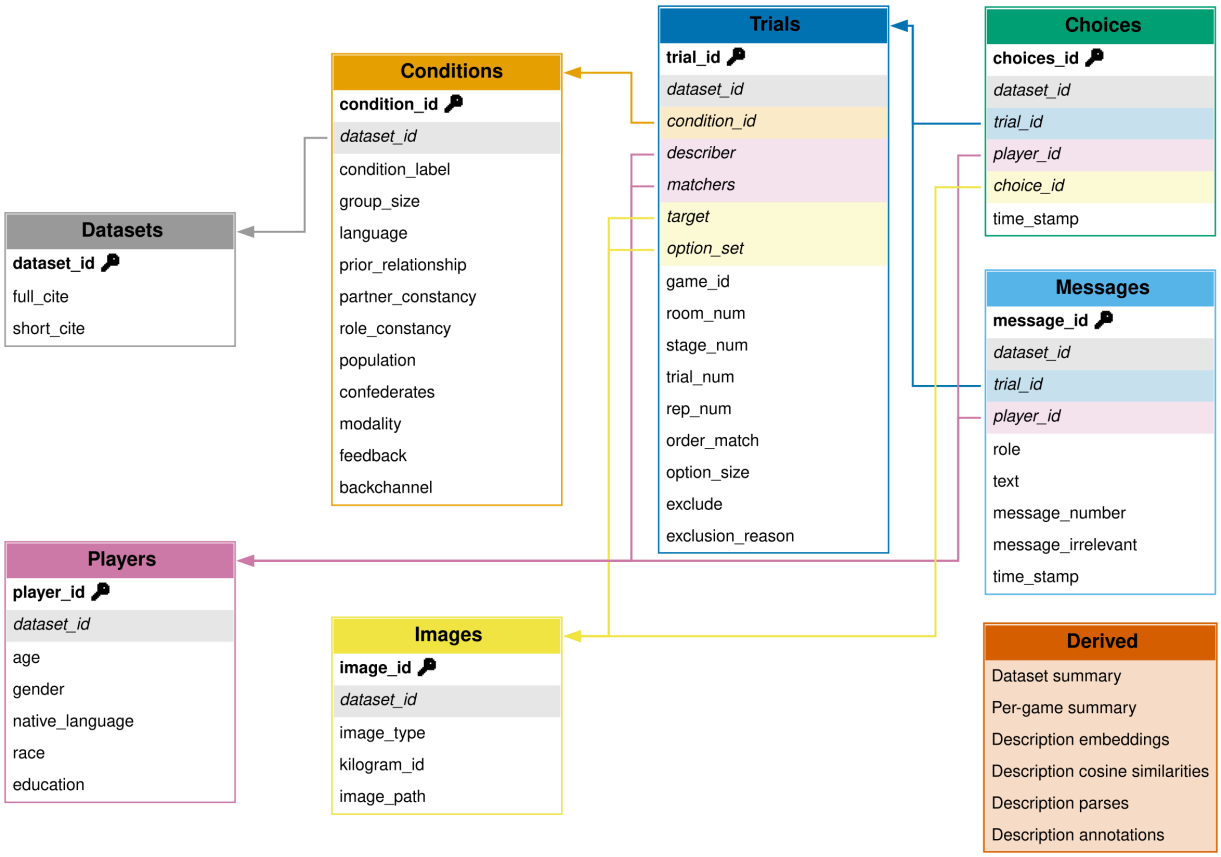


Figure 1. Schematic illustration of the different tables containing different pieces of data about the iterated reference games and how they connect. Arrows show connections between rows that key into other tables. Redundant connections back to the dataset table are not shown.

they are active; roles are not constant if a matcher becomes a describer or vice versa.

- backchannel: designs vary on how the matchers may communicate with the describer. We break this up into 3 levels: full if there are no restrictions on communication (within the modality), limited if there is some communication (example, if only one message can be sent), and none if there is no communication channel from matcher to describer.
- feedback: we use "feedback" to refer to information players receive on the correctness of their selections from the experimenters. In full feedback, after each trial, players learn if the selection(s) were correct or incorrect and what the correct target was. In limited feedback, there is some correctness feedback, but it doesn't include what the correct target was, or is provided after every repetition, perhaps as a total correct. In no feedback, players receive no feedback during the game.
- modality: We code the communication modality into three levels: oral-in-person when the participants are talking and physically in the same room, whether or not they have visual access to each other; oral-remote when the participants are talking over some sort of connection (audio signal between phone booths, headsets in separate fMRI machines, etc). We use written when the messages are typed, like via a chatbox.

- confederates: we include an option for noting if one or more of the participants was not a "real" participant – i.e. was an experimenter or a bot. Iterated reference games involving some artificial intelligence agents are a topic of recent study, and Refbank is ready to include those studies, and also mark them so then can be treated separately or excluded as appropriate for the research goals.
- prior-relationship: another possible source of design variation is whether the pair or group of participants have prior history with each other, which could shape their interactions and convention formation. In some cases, the populations were mixed, either with an explicit balance between familiar pairs or stranger pairs, or because the recruitment pool included some classmates and some strangers.

Players. The player table contains some basic demographic information about participants when available. It can be linked via the player IDs. We track players' age, gender, native language, race, and level of education, although these are not always available across datasets.

Images. Each image has a unique image ID for cross referencing with the trials and choices tables. We also record image type (tangram, line drawing, fribble, etc). For tangram images that are in the Kilogram dataset (Ji et al., 2022), we include the Kilogram ID for cross-referencing. When we have access to the actual image file, we store these and include their image path. The same images are often reused across different datasets, and the image records are deduplicated so multiple papers may refer to the same image ID.

Derived

The core and metadata tables represent the raw data that we import from individual studies. We also have a workflow for processing the raw data and creating derivative measures that may be commonly of use to researchers. These derivative tables are created via a Redivis workflow. For example, word counts can be derived from the text data in messages but are a common enough metric that we calculate them for a summary table. Other derived tables contain vector embeddings of the messages from Jina AI embeddings (Akram et al., 2026), cosine similarities between the embeddings, and Stanza parses of the messages (Qi et al., 2020). In producing these measures, we apply our own light exclusions, requiring that the describer did produced descriptions, and that there are at least 2 "blocks" worth of trials for that game. As additional processing pipelines are developed, we may add more derived tables.

Processing

We started developing Refbank with the iterated reference games that were conducted by members of the team, or where the data was already posted publicly. We then contacted authors of a number of iterated reference game papers asking if they were willing to share their data.

We received data in a variety of formats, including pre-cleaned tabular data, output from experimental software like Empirica (Almaatouq et al., 2021), transcripts, and raw audio files. We transcribed raw audio using WhisperX (large-v3, Bain et al., 2023). When transcripts or transcribed audio did not have trial boundaries, we hired a contractor to annotate trial boundaries by hand (and confirmed high agreement with an author) or

annotated trial boundaries by hand. When target to image correspondence was not provided, we marked the correspondences we thought were most likely. In the import repository, each dataset has a README including citation information, the abstract, the key facts about the study design, and notes about any inferences or processing we had to do as part of the import process.

Datasets also varied in completeness; some datasets had timing information and demographics, whereas in others only the transcripts were available, or only selection data. We have imported what was available, but due to incomplete data, some datasets might not be applicable for all analyses.

We imported trials from all games, including games that were excluded in the initial studies. We mark their exclusion status and reason, but allow users of Refbank to impose their own uniform exclusion criteria.

Metadata was coded from the data and from the paper or write-up. In a few cases, condition information was not fully specified in the paper. In these cases, we either talked with the authors, or made our best guesses from the paper. When there were borderline cases that were not explicit we inferred them and made a note in the README.

Some of the contributed datasets had additional information beyond the shared metrics. We do not include these additional data (such as pre- or post- tests, mouse or eye tracking, fMRI data, or audio files).

Current datasets

Refbank currently has data from 19 datasets including 1,847,175 words from 146,579 referential trials from 2,726 games and 8,098 participants (Table 2).

These datasets are primarily in English, although Refbank includes one dataset each in French, Swiss German, and Dutch, and one dataset with codeswitching between Spanish and English. Datasets are primarily made up of adult participants, although 3 datasets include data from children: two of just children interacting with peers (Boyce et al., 2026; Branigan et al., 2016), and one of child–parent pairs (Leung et al., 2024). Datasets vary in their design features (Table 1). These dataset characteristics reflect both the distribution of papers published and what datasets were made available to Refbank. We welcome any additional datasets, including unpublished datasets.

[unclear where to stick this] Transcripts, selections, and metadata are not human subjects data under ... [TODO someone write this], and so likely can be shared. (could crib from the peekbank behavioral methods paper for how they framed this!)

Using Refbank

Refbank data is stored on Redivis, a platform for hosting scientific data. A Redivis account can be made freely and is used to access the data. Data is versioned, so that either the current version or a previous version of data can be used (for reproducibility).

There are several ways of interacting with the Refbank data. We recommend using the observable application for quick visualizations and exploration of moderators or different datasets, via the Refbank website (refbank.stanford.edu). For analyzing data, the data tables can be downloaded via the Redivis API for Python, or using `refbankr`, a custom R package.

Table 1
Design features of each dataset. 'ltd' is used as an abbrevation of limited. Multiple values indicate that the paper reported on different conditions that varied in the given feature.

Paper	Language	Modality	Role switch	Partner switch	feedback	backchannel
Bangerter & Smokenski (2000)	Swiss German	oral-in-person	X		none	full
Bangerter et al. (2020)	French	oral-in-person		mixed	none	full
Beatty-Martinez et al. (2026)	Spanish/English	written	X		full	full
Bögels et al. (2026)	Dutch	oral-in-person	X		none	full
Boyce et al. (2024)	English	written	mixed		ltd, full	ltd, full
Boyce et al. (2026)	English	oral-in-person	X		full	full
Branigan et al. (2016)	English	oral-in-person		X	ltd	full
Dahan (2023)	English	oral-in-person	X		none	full
Dale et al. (2011)	English	oral-remote			none	full
Hawkins et al. (2019)	English	written			full	none
Hawkins et al. (2020)	English	written			full	full
Hawkins et al. (2021)	English	written	X	X	full	full
Hawkins et al. (2023)	English	written	X	X	full	full
Hawkins et al. (2026)	English	oral-in-person	X		full	full
Ji et al. (2025)	English	written	X		ltd	none, full
Leung et al. (2024)	English	oral-in-person	X		none	full
Mankewitz & Hawkins (2025)	English	written	X		full	full
Wang et al. (2025)	English	oral-remote			none	full
Yoon & Brown-Schmidt (2019)	English	oral-in-person		X	none	full

`refbankr` functions can be used to pull any set of datasets, any version, and also to pull a sample of the data for checks.

Redivis has built-in methods of filtering, exploring, and downloading data, via its viewers and workflow creation, where a chain of SQL transforms, and R and Python notebooks can be used to process data. Redivis allows for the creation of workflows that pull from datasets, and for SQL-type queries over datasets.

Different workflows support different possible uses including scientific research and training. The API format makes them usable including in larger pipelines and is compatible with coding agents [cite anthropic blogpost?].

Analyses

We show a few illustrative analyses of Refbank data to illustrate the use of aggregated data. For these analyses, we only considered games and targets where the target was referred to at least 3 times during the game and where message data was available.

We also restrict to the first stage of each experiment; this is the part of the experiment before any changes in partners or the set of stimuli being used. Thus, for datasets like Hawkins, Goodman, et al. (2020), even though the overall design has games of 4 people interacting as a network of pairs, for the first stage, each game can be treated as two independent pairs. Similarly, games that eventually switch for a new set of stimuli look like typical games without stimuli swaps for the repetitions before that happens.

Table 2
Amount of data from each dataset, and the amount of data per game in each dataset. Ranges indicate that the value varied across conditions within the dataset. Options per trial in parentheses indicate that a minority of trials has this other number of image options.

Paper	Total games	Total players	Total trials	Total words	Trials/ game	Reps/ game	Group size	Option per trial
Yoon & Brown-Schmidt (2019)	788	3 614	41 766	348 849	80 - 208	5 - 13	4 - 7	16 (4)
Mankewitz & Hawkins (2025)	447	897	28 244	198 713	64	4	2	4
Boyce et al. (2024)	306	1 255	21 269	374 105	72	6	2 - 6	12
Ji et al. (2025)	262	524	8 183	50 468	30 - 40	5 - 6	2	10
Hawkins et al. (2020)	173	346	12 376	128 274	72	6	2	12
Hawkins et al. (2019)	168	279	3 994	19 740	24	6	2	4
Boyce et al. (2026)	102	199	1 513	17 661	12 - 16	3 - 4	2	2
Leung et al. (2024)	82	166	3 259	37 360	40	4	2	2
Bangerter et al. (2020)	76	20	3 757	161 149	40 - 64	5 - 8	2 - 3	8
Bögels et al. (2026)	47	95	4 512	176 483	96	6	2	16
Beatty-Martinez et al. (2026)	45	91	2 698	30 848	60	5	2	12
Hawkins et al. (2021)	43	172	3 699	25 777	96	12	4	4
Hawkins et al. (2026)	42	86	3 761	96 102	180	8	2	18 (36)
Dahan (2023)	40	80	1 920	96 368	48	16	2	16
Hawkins et al. (2023)	26	105	2 278	11 756	96	12	4	4
Branigan et al. (2016)	24	74	764	30 430	32	4	3	8
Dale et al. (2011)	20	40	359	0	18	3	2	6
Wang et al. (2025)	20	21	800	26 105	40	4	2	10
Bangerter & Smokenski (2000)	15	34	1 427	16 987	96	12	2	8
Total	2 726	8 098	146 579	1 847 175				

Accuracy

A common finding across iterated reference games is that players are usually quite accurate at communicating and selecting the correct targets – that is, they succeed at reference. When participants are not at ceiling initially, they tend to improve rapidly over repetition. A high level of accuracy is taken as a pre-condition for interpreting the language data as being a sign of conventionalization.

Not all datasets included accuracy data on the trial level, so these results come from the 14 datasets (131,484 trials) where selection data and text data were available.

To aggregate across datasets, we ran a mixed-effect logistic regression model correct$\sim$ log rep num $\times$ option size + (log rep num||dataset/condition). The fitted results are shown in Figure 2A. Consistent with previous findings, accuracy is generally high to start with, but also increases over repetitions (odds ratio of one log-rep-num later 1.96 [1.32, 2.80]).

Reduction of utterance length

A very consistent finding across many iterated reference games is that the amount of language used in descriptions declines across repetitions. However, while the trend is frequently observed, the specific mechanisms are not known, and the functional relationship between repetition and quantity of words has not been firmly established. Bögels et al. (2026) compared a linear model and a log-log (power law) model on two datasets (Eijk et al., 2022; Hawkins, Frank, et al., 2020), and found that log-log model provided a better fit to the

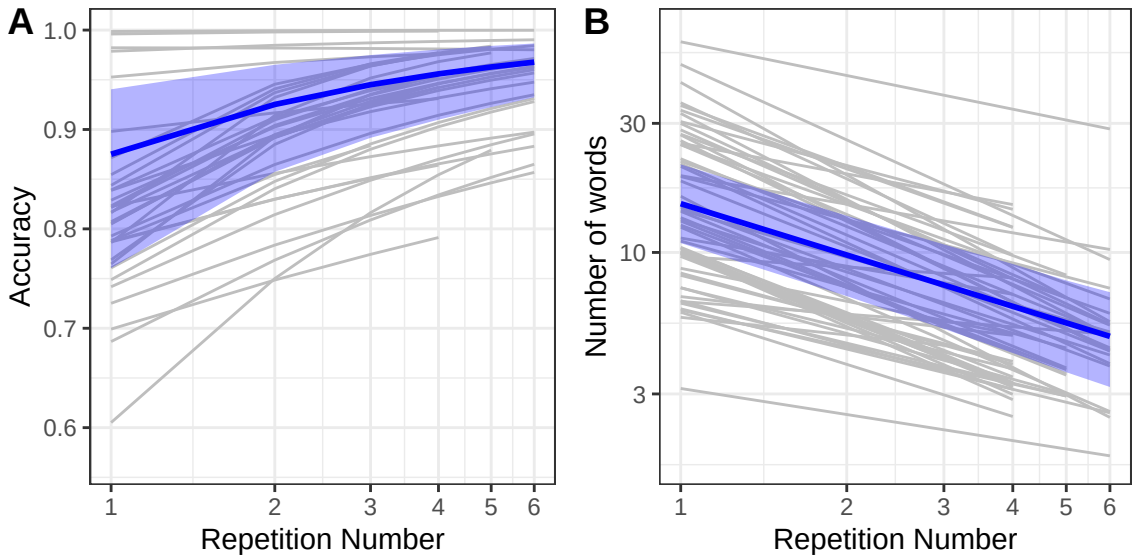


Figure 2. Model predictors for accuracy (A) and describer utterance length in number of words(B). Each condition level estimate is shown in a faint line, and the generalized prediction is shown in blue, with a 95% CI.

data.

While there are many possible functional relationships between repetition and number of words, here we consider four options in order to determine what general type of shape fits the data best. By using a larger number of datasets that vary in different ways, we avoid overfitting to the pecularities of any one dataset.

We consider relationships between repetition and words on either the raw or log scales for each. For each model we included mixed effects (e.g., log words∼ rep num + (rep num||dataset/condition)). We found that the best fitting model was log(words)~log(rep), but it was only slightly better than log(words)~rep (ELPD difference of -1327). Models with a linear number of words were much worse (ELPD differences of -103,214 and -105,580).

Using the best fitting log-log model, we confirm the common finding that there is a substantial decrease in the number of words over repetitions (Figure 2B). There was a change of -0.63 [-0.76, -0.51] log words per log repetition number.

Reduction moderators

One analysis enabled by the aggregation of data across studies is the examination of the possible effects of experiment design features on the rate of reduction – that is, whether some conditions result in faster reduction than others. For this analysis, we used the log(words)~log(rep) model and extracted the random slopes for log-rep-num for each condition. We then predicted the slope value as a function of experiment design features: number of players, option set size, image type, partner constancy, role constancy, population (adults vs children), modality, availability of feedback, and availability of matcher backchanneling. Because of the large number of possible moderators, we used XGBoost, a gradient-boosted tree approach to prediction. This method involves fitting a large number of decision trees based on the values of the predictors to predict the by-condition slopes,

thereby allowing arbitrary non-linearities and interactions among the predictors.

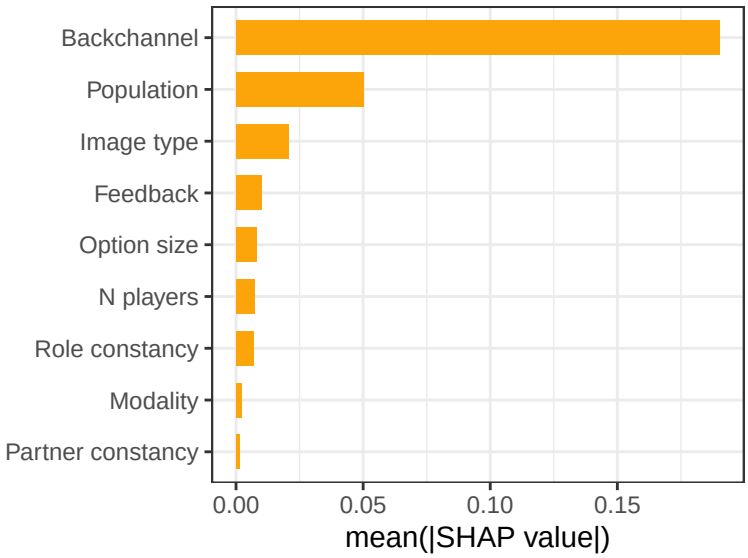


Figure 3. Mean absolute SHAP values for experiment design features on the slope of reduction. Higher values indicate greater influence on slope predictions.

Figure 3 shows the mean absolute SHAP values for experiment design features on the slope of reduction. The SHAP values are an indicator of the importance of each predictor, such that higher values reflect a greater influence of that variable on the predicted value. The most influential feature by far was backchanneling, such that full backchanneling resulted in faster reduction; this result reflects the importance of bidirectional communication on converging to a common ground. Population and image type were also important predictive features, and all other features had close to no effect on the predictions.

Semantic embedding similiarities

One way of quantifying the semantic trajectories of descriptions over repetitions is to embed the descriptions in a semantic vector space and then quantify similarities of pairs of descriptions using cosine similarity or another distance metric. This approach is used in analyses of dialogue trajectories [Reece et al. (2023); Fischer and Ram (2024); CITES].

As part of the Refbank processing pipeline, we embed all descriptions using a multilingual text embedding model, specifically jina-embeddings-v5-text-small (Akram et al., 2026; Feng et al., 2022). We then calculate cosine similarities between relevant pairs of descriptions to measure how the semantic distances of descriptions change across repetition number.

Within a game, the similarity between descriptions to the same target in adjacent repetitions is an index of convention formation (Boyce et al., 2024; Hawkins, Frank, et al., 2020). When there is low similarity, descriptions are changing, but when there is high similarity the target to description mapping has stabilized. To test for stabilization of descriptions, we ran a mixed effects ordered beta regression model sim$\sim$ log rep num + (log rep num||dataset/condition) (Kubinec, 2023). In this model, the similarity between a description and the corresponding description in the next repetition increases over

repetitions by 0.37 [0.26, 0.46] per log repetition. The predicted trajectories for a generic reference game and each individual condition are shown in Figure 4A.

As games converge to a variety of different, arbitrary conventionalized names, these descriptions often become differentiated from one another (Boyce et al., 2024; Hawkins, Frank, et al., 2020). To test for divergence between different games, we ran a mixed effects ordered beta regression model sim∼ log rep num + (log rep num||dataset/condition) predicting the average distance of a description to other descriptions for the same repetition and target, but a different game in the same condition. In this model, the similarity between descriptions from different games changes by -0.17 [-0.22, -0.12] per log repetition. The predicted trajectories for a generic reference game and each individual condition are shown in Figure 4B.

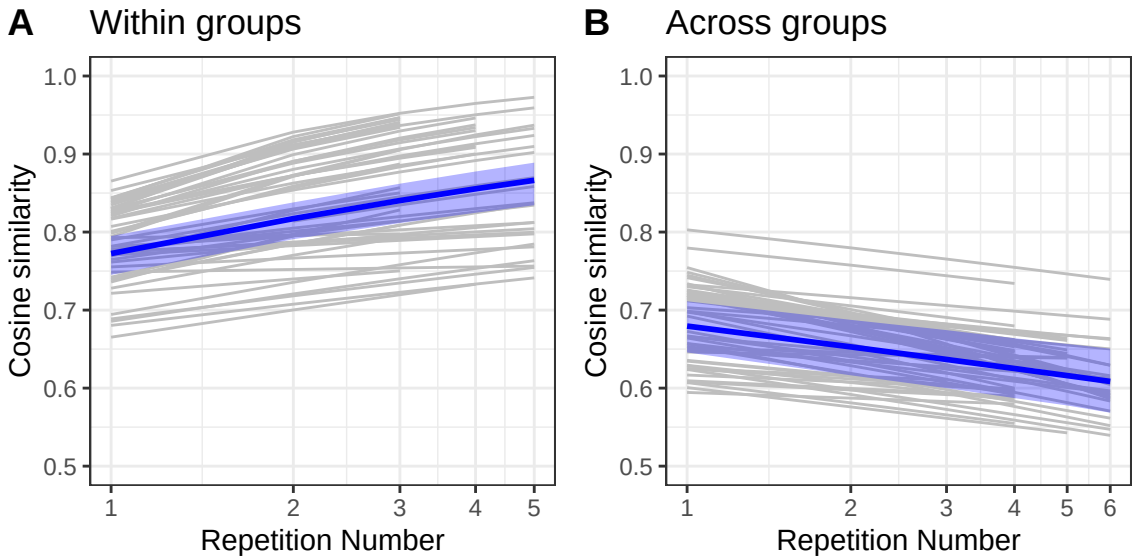


Figure 4. Cosine similarities between a description and the next repetition description from the same game (A) or descriptions from the same repetition and different games (B). Blue line shows the overall model prediction and 95% CrI, faint lines show per-condition predictions.

Distribution of parts of speech

The distributions of words and phrases used in descriptions changes as conventionalization occurs. For instance, definite or bare noun phrases may increase compared to indefinites (Brown-Schmidt & Duff, 2016; Dahan, 2025; Isaacs & Clark, 1987), and more holistic or metaphorical descriptions may become more common relative to literal, shape-based, or part-based descriptions (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986). As as illustration of how Refbank and natural language processing pipelines can be used to scale up reference game analyses, we focus on the change in the distribution of parts of speech. Hawkins, Frank, et al. (2020) found that the distribution of parts of speech in describer utterances changes across repetitions, with nouns becoming relatively more common at the expense of closed-class words. We are able to scale up this analysis to all the monolingual datasets in Refbank. We use Stanza to parse each message (Qi et al., 2020). The part of speech distributions and full dependency parses are saved in derived tables in Refbank.

For this analysis, we lump parts of speech into 6 coarse grained categories: nouns, verbs, modifiers (adjectives and adverbs), pronouns, determiners, and function words (adpositions, conjunctions, etc). We ran a mixed effects multinomial model with pronoun as the reference class, weighting each trial equally: cbind(noun,verb,modifier,function,det,pron)|trials(total) + weights(w)∼ log rep num × modality + (log rep num||dataset/condition). After initial visualization suggested that spoken and written corpora might vary, we added modality (oral or written) as a predictor in an exploratory model. According to the model predictions, from the first to sixth repetition, the fraction of nouns is expected to increase 0.15 [0.05, 0.27], while function words (-0.07 [-0.12, -0.01]) and pronouns decrease in frequency (-0.05 [-0.06, -0.04]). Other categories were less consistent. Modality makes the largest difference in the size of the noun boost (0.11 [0.02, 0.21] for spoken versus 0.20 [0.07, 0.32] for written) and the direction of change for determiners, where determiner rate is neutral to increasing in spoken corpora (0.05 [-0.02, 0.14]) and decreasing in written corpora (-0.09 [-0.13, -0.04]). The model predicted results from the model with modality are shown in Figure 5.

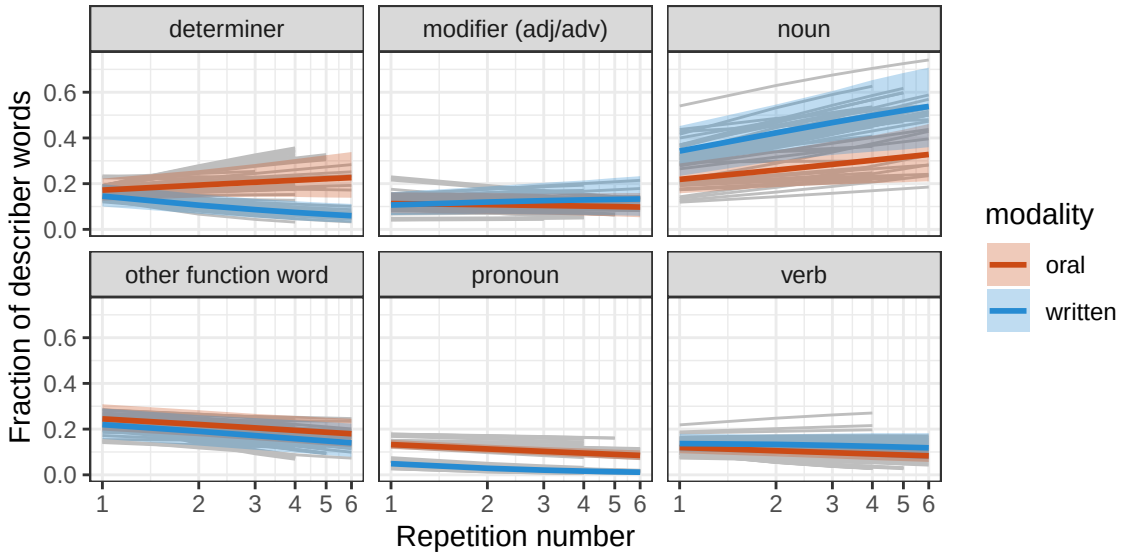


Figure 5. Distribution in parts of speech over repetitions. Colored lines and ribbons are the predictions and 95%CrI from the model; faint lines are per-condition model predictions.

Discussion

Iterated reference games are a classic paradigm that has been used to study numerous aspects of dialogue and communication, and continues to be used in new ways, including as a way of assessing vision–language models. While iterated reference games share some core properties, games can also vary along a number of design parameters. In addition, iterated reference games are analysed in a number of different ways, with more datasets being studied on a subset of possible questions. Even for research questions that are commonly reported on, such as change in number of words over repetition, differences in design and modeling approach and reporting make it difficult to quantitatively synthesis results. Taken together, these factors mean that iterated reference games are a good candidate for a harmonized

repository: data can be unified, data have high reuse value, and alternative synthesis methods such as meta-analysis are not as suitable.

In this paper, we introduced Refbank, an open repository of iterated reference game data. We then presented a series of analyses using Refbank data, many of which would not have been feasible without the raw data from different experiments.

The functional relationship between utterance length and repetition is more robustly fit over a range of datasets, and we provide additional evidence that the relationship is roughly log-log or log-linear, confirming Bögels et al. (2026). Moreover, using the coded metadata, we were able to examine which design features have the largest impact on the slope of reduction. Knowing whether certain design features matter is important to experiment design, but requires large amounts and diversity of data to study.

Semantic trajectories have been analysed in iterated reference games before, but by scaling up a unified pipeline, we are able to analyse multiple datasets, including in different languages, in the same pipeline. We confirm that the trends of within game semantic convergence and between game semantic divergence, that had previously been observed in a few written-modality tangram studies extend to spoken studies, studies using other stimuli, and studies in different languages.

The possibility for cross-dataset comparisons also allows Refbank to enable novel findings: in an analysis of distributions of part of speech across repetition, we found some divergences between spoken and typed paradigms, where noun proportion increased more sharply in typed datasets, and determiners increased in spoken datasets, but not written ones. This may reflect differences in the cost to produce different parts of speech in different modalities. A more telegraphic approach may be easier in typing where cost is per character, but in speaking, determiners and other short function words may be faster to say than to omit. This idea of differential cost based on modality suggests a direction for future work that more closely manipulates modality and cost in controlled, comparable experiments.

These analyses are a gesture towards what Refbank makes possible. By unifying datasets in a common framework, we make possible generalization of existing analyses to a larger collection of datasets, the use of new processing pipelines on the data as techniques arise, and the study of variation across datasets and design choices. The dataset is set-up for compatibility with natural language processing tools as pipelines can be created.

Limitations

While Refbank provides size and diversity compared to any individual study, Refbank's generalizability is limited by the datasets that are collected and the datasets that are shared with Refbank. Some papers that collected reference game data no longer have their raw data or were unwilling to share. Additionally, most iterated reference games are conducted in English, reflected in the linguistic skew of Refbank. A broader distribution of languages is necessary for some questions around how language changes in a reference game.

While Refbank has these limitations, it can also be used to identify gaps in the current pool of reference games and point the way to future high value experiments. The data are heterogenous, but metadata is coded, so researchers can filter to the appropriate datasets, games, and trials for their analyses.

Conclusion

We encourage the sharing of iterated reference game data and hope that Refbank will house more games in the future.

Refbank can be used to study the moderators and generalizations of convention formation phenomena in iterated reference games, and identify gaps where future experiments could resolve theoretical questions. Refbank is also a corpus of task-specific dialogues that can be used for corpus linguistics, for assessing pragmatic phenomena in language models, and as a pedagogical resource for instructors and students alike.

References

Akram, M. K., Sturua, S., Havriushenko, N., Herreros, Q., Günther, M., Werk, M., & Xiao, H. (2026). *Jina-embeddings-v5-text: Task-Targeted Embedding Distillation* (arXiv:2602.15547). arXiv. https://doi.org/10.48550/arXiv.2602.15547

Almaatouq, A., Becker, J., Houghton, J. P., Paton, N., Watts, D. J., & Whiting, M. E. (2021). Empirica: A virtual lab for high-throughput macro-level experiments. *Behavior Research Methods*, *53*(5), 2158–2171. https://doi.org/10.3758/s13428-020-01535-9

Bain, M., Huh, J., Han, T., & Zisserman, A. (2023). *WhisperX: Time-Accurate Speech Transcription of Long-Form Audio* (arXiv:2303.00747). arXiv. https://doi.org/10.48550/arXiv.2303.00747

Bangerter, A., Mayor, E., & Knutsen, D. (2020). Lexical entrainment without conceptual pacts? Revisiting the matching task. *Journal of Memory and Language*, *114*, 104129. https://doi.org/10.1016/j.jml.2020.104129

Bögels, S., Li, T., Rasenberg, M., Eijk, L., Toni, I., & Pouw, W. (2026). There is a power law of joint communicative effort and it reflects communicative work. *Cognition*, *268*, 106370. https://doi.org/10.1016/j.cognition.2025.106370

Bovet, V., Knutsen, D., & Fossard, M. (2023). Direct and indirect linguistic measures of common ground in dialogue studies involving a matching task: A systematic review. *Psychonomic Bulletin &Amp; Review*, *31*(1), 122–136. https://doi.org/10.3758/s13423-023-02359-2

Boyce, V., Hawkins, R. D., Goodman, N. D., & Frank, M. C. (2024). Interaction structure constrains the emergence of conventions in group communication. *Proceedings of the National Academy of Sciences*, *121*(28), e2403888121. https://doi.org/10.1073/pnas.2403888121

Boyce, V., Sparks, R. Z., & Frank, M. C. (2026). *Preschoolers can coordinate with each other to communicate about novel referents* (t76a5_v2). PsyArXiv.

Branigan, H. P., Bell, J., & McLean, J. F. (2016). Do You Know What I Know? The Impact of Participant Role in Children's Referential Communication. *Frontiers in Psychology*, *7*. https://doi.org/10.3389/fpsyg.2016.00213

Brennan, S. E., & Clark, H. H. (1996). Conceptual pacts and lexical choice in conversation. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *22*(6), 1482–1493. https://doi.org/10.1037/0278-7393.22.6.1482

Brown-Schmidt, S., & Duff, M. C. (2016). Memory and Common Ground Processes in Language Use. *Topics in Cognitive Science*, *8*(4), 722–736. https://doi.org/10.1111/tops.12224

Clark, H. H., & Wilkes-Gibbs, D. (1986). Referring as a collaborative process. *Cognition*, *22*(1), 1–39. https://doi.org/10.1016/0010-0277(86)90010-7

Dahan, D. (2024). When hedging helps, rather than impedes, communication: Collaboration in the referential communication task. *Discourse Processes*, *62*(2), 89–111. https://doi.org/10.1080/0163853X.2024.2437738

Dahan, D. (2025). Coordinating reference in conversation: The choice between linguistic conventions and linguistic precedents. *Journal of Memory and Language*, *143*, 104639. https://doi.org/10.1016/j.jml.2025.104639

Eijk, L., Rasenberg, M., Arnese, F., Blokpoel, M., Dingemanse, M., Doeller, C. F., Ernestus,

M., Holler, J., Milivojevic, B., Özyürek, A., Pouw, W., van Rooij, I., Schriefers, H., Toni, I., Trujillo, J., & Bögels, S. (2022). The CABB dataset: A multimodal corpus of communicative interactions for behavioural and neural analyses. *NeuroImage*, *264*, 119734. https://doi.org/10.1016/j.neuroimage.2022.119734

Eliav, R., Ji, A., Artzi, Y., & Hawkins, R. D. (2023). *Semantic uncertainty guides the extension of conventions to new referents* (arXiv:2305.06539). arXiv. https://doi.org/10.48550/arXiv.2305.06539

Feng, F., Yang, Y., Cer, D., Arivazhagan, N., & Wang, W. (2022). Language-agnostic BERT Sentence Embedding. In S. Muresan, P. Nakov, & A. Villavicencio (Eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 878–891). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.acl-long.62

Fischer, J. R., & Ram, N. (2024). Personality Differences Drive Conversational Dynamics: A High-Dimensional NLP Approach. In J. Hale, K. Chawla, & M. Garg (Eds.), *Proceedings of the Second Workshop on Social Influence in Conversations (SICon 2024)* (pp. 36–45). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.sicon-1.3

Fox Tree, J. E., & Clark, N. B. (2013). Communicative Effectiveness of Written Versus Spoken Feedback. *Discourse Processes*, *50*(5), 339–359. https://doi.org/10.1080/0163853X.2013.797241

Frank, M. C., Braginsky, M., Yurovsky, D., & Marchman, V. A. (2017). Wordbank: An open repository for developmental vocabulary data. *Journal of Child Language*, *44*(3), 677–694.

Frank, M. C., Braginsky, M., Yurovsky, D., & Marchman, V. A. (2021). *Variability and consistency in early language learning: The wordbank project*. MIT Press.

Ghaleb, E., Rasenberg, M., Pouw, W., Holler, J., & Fernandez, R. (2024). *Analysing Cross-Speaker Convergence in Face-to-Face Dialogue through the Lens of Automatically Detected Shared Linguistic Constructions*. https://doi.org/10.48550/arXiv.2405.08546

Hawkins, R. D., Frank, M. C., & Goodman, N. D. (2020). Characterizing the Dynamics of Learning in Repeated Reference Games. *Cognitive Science*, *44*(6). https://doi.org/10.1111/cogs.12845

Hawkins, R. D., Goodman, N. D., Goldberg, A. E., & Griffiths, T. L. (2020). Generalizing meanings from partners to populations: Hierarchical inference supports convention formation on networks. *arXiv:2002.01510 [Cs]*. https://doi.org/10.48550/arxiv.2002.01510

Hawkins, R. D., Liu, I., Goldberg, A. E., & Griffiths, T. G. (2021). Respect the code: Speakers expect novel conventions to generalize within but not across social group boundaries. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *43*.

Horton, W. S., & Gerrig, R. J. (2002). Speakers' experiences and audience design: Knowing when and knowing how to adjust utterances to addressees. *Journal of Memory and Language*, *47*(4), 589–606. https://doi.org/10.1016/S0749-596X(02)00019-0

Horton, W. S., & Gerrig, R. J. (2005). The impact of memory demands on audience design during language production. *Cognition*, *96*(2), 127–142. https://doi.org/10.1016/j.cognition.2004.07.001

Horton, W. S., & Spieler, D. H. (2007). Age-related differences in communication and audience design. *Psychology and Aging*, *22*(2), 281–290.

https://doi.org/10.1037/0882-7974.22.2.281
Hupet, M., Chantraine, Y., & Nef, F. (1993). References in conversation between young and old normal adults. *Psychology and Aging*, *8*(3), 339–346. https://doi.org/10.1037/0882-7974.8.3.339
Isaacs, E. A., & Clark, H. H. (1987). References in conversation between experts and novices. *Journal of Experimental Psychology: General*, *116*(1), 26–37. https://doi.org/10.1037/0096-3445.116.1.26
Ji, A., Kojima, N., Rush, N., Suhr, A., Vong, W. K., Hawkins, R., & Artzi, Y. (2022). Abstract Visual Reasoning with Tangram Shapes. In Y. Goldberg, Z. Kozareva, & Y. Zhang (Eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (pp. 582–601). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.emnlp-main.38
Knutsen, D., & Le Bigot, L. (2018). The influence of conceptual (mis)match on collaborative referring in dialogue. *Psychological Research*, *84*(2), 514–527. https://doi.org/10.1007/s00426-018-1060-1
Koile, E., & Cristia, A. (2021). Toward Cumulative Cognitive Science: A Comparison of Meta-Analysis, Mega-Analysis, and Hybrid Approaches. *Open Mind*, *5*, 154–173. https://doi.org/10.1162/opmi_a_00048
Krauss, R. M., & Bricker, P. D. (1967). Effects of Transmission Delay and Access Delay on the Efficiency of Verbal Communication. *The Journal of the Acoustical Society of America*, *41*(2), 286–292. https://doi.org/10.1121/1.1910338
Krauss, R. M., & Glucksberg, S. (1977). Social and Nonsocial Speech. *Scientific American*, *236*(2), 100–105. https://doi.org/10.1038/scientificamerican0277-100
Krauss, R. M., & Weinheimer, S. (1964). Changes in reference phrases as a function of frequency of usage in social interaction: A preliminary study. *Psychonomic Science*, *1*(1-12), 113–114. https://doi.org/10.3758/BF03342817
Krauss, R. M., & Weinheimer, S. (1966). Concurrent feedback, confirmation, and the encoding of referents in verbal communication. *Journal of Personality and Social Psychology*, *4*(3), 343–346. https://doi.org/10.1037/h0023705
Kubinec, R. (2023). Ordered Beta Regression: A Parsimonious, Well-Fitting Model for Continuous Data with Lower and Upper Bounds. *Political Analysis*, *31*(4), 519–536. https://doi.org/10.1017/pan.2022.20
Leung, A., Yurovsky, D., & Hawkins, R. D. (2024). Parents spontaneously scaffold the formation of conversational pacts with their children. *Child Development*, *96*(2), 546–561. https://doi.org/10.1111/cdev.14186
Lysander, K., & Horton, W. S. (2012). Conversational Grounding in Younger and Older Adults: The Effect of Partner Visibility and Referent Abstractness in Task-Oriented Dialogue. *Discourse Processes*, *49*(1), 29–60. https://doi.org/10.1080/0163853X.2011.625547
MacWhinney, B. (2000). *The CHILDES project: Tools for analyzing talk*. Lawrence Erlbaum Associates.
Markiewicz, C. J., Gorgolewski, K. J., Feingold, F., Blair, R., Halchenko, Y. O., Miller, E., Hardcastle, N., Wexler, J., Esteban, O., Goncavles, M., Jwa, A., & Poldrack, R. (2021). The OpenNeuro resource for sharing of neuroscience data. *eLife*, *10*, e71774. https://doi.org/10.7554/eLife.71774

Metzing, C., & Brennan, S. (2003). When conceptual pacts are broken: Partner-specific effects on the comprehension of referring expressions. *Journal of Memory and Language*, *49*(2), 201–213. https://doi.org/10.1016/S0749-596X(03)00028-7

Qi, P., Zhang, Y., Zhang, Y., Bolton, J., & Manning, C. D. (2020). *Stanza: A Python Natural Language Processing Toolkit for Many Human Languages* (arXiv:2003.07082). arXiv. https://doi.org/10.48550/arXiv.2003.07082

Reece, A., Cooney, G., Bull, P., Chung, C., Dawson, B., Fitzpatrick, C., Glazer, T., Knox, D., Liebscher, A., & Marin, S. (2023). The CANDOR corpus: Insights from a large multimodal dataset of naturalistic conversation. *Science Advances*, *9*(13), eadf3197. https://doi.org/10.1126/sciadv.adf3197

Schober, M. F., & Clark, H. H. (1989). Understanding by addressees and overhearers. *Cognitive Psychology*, *21*(2), 211–232. https://doi.org/10.1016/0010-0285(89)90008-X

Turner, C., & Knutsen, D. (2021). Audience Design in Collaborative Dialogue between Teachers and Students. *Discourse Processes*, *58*(8), 703–725. https://doi.org/10.1080/0163853X.2021.1904768

Wilkes-Gibbs, D., & Clark, H. H. (1992). Coordinating beliefs in conversation. *Journal of Memory and Language*, *31*(2), 183–194. https://doi.org/10.1016/0749-596X(92)90010-U

Yoon, S. O., & Brown-Schmidt, S. (2014). Adjusting conceptual pacts in three-party conversation. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *40*(4), 919–937. https://doi.org/10.1037/a0036161

Yoon, S. O., & Brown-Schmidt, S. (2019a). Audience Design in Multiparty Conversation. *Cognitive Science*, *43*(8), e12774. https://doi.org/10.1111/cogs.12774

Yoon, S. O., & Brown-Schmidt, S. (2019b). Contextual Integration in Multiparty Audience Design. *Cognitive Science*, *43*(12), e12807. https://doi.org/10.1111/cogs.12807

Zettersten, M., Yurovsky, D., Xu, T. L., Uner, S., Tsui, A. S. M., Schneider, R. M., Saleh, A. N., Meylan, S. C., Marchman, V. A., Mankewitz, J., MacDonald, K., Long, B., Lewis, M., Kachergis, G., Handa, K., deMayo, B., Carstensen, A., Braginsky, M., Boyce, V., … Frank, M. C. (2022). Peekbank: An open, large-scale repository for developmental eye-tracking data of children's word recognition. *Behavior Research Methods*, *55*(5), 2485–2500. https://doi.org/10.3758/s13428-022-01906-4